

REVIEW

Unifying individual differences in personality, predictability and plasticity: A practical guide

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Australian Research Council, Grant/Award Number: DE150101774 and DP180100818; Australian Government Research Training Program Scholarship

Handling Editor: Robert Freckleton**Abstract**

1. Organisms use labile traits to respond to different conditions over short time-scales. When a population experiences the same conditions, we might expect all individuals to adjust their trait expression to the same, optimal, value, thereby minimising phenotypic variation. Instead, variation abounds. Individuals substantially differ not only from each other, but also from their former selves, with the expression of labile traits varying both predictably and unpredictably over time.
2. A powerful tool for studying the evolution of phenotypic variation in labile traits is the mixed model. Here, we review how mixed models are used to quantify individual differences in both means and variability, and their between-individual correlations. Individuals can differ in their average phenotypes (e.g. behavioural personalities), their variability (known as 'predictability' or intra-individual variability), and their plastic response to different contexts.
3. We provide detailed descriptions and resources for simultaneously modelling individual differences in averages, plasticity and predictability. Empiricists can use these methods to quantify how traits covary across individuals and test theoretical ideas about phenotypic integration. These methods can be extended to incorporate plastic changes in predictability (termed 'stochastic malleability').
4. Overall, we showcase the unfulfilled potential of existing statistical tools to test more holistic and nuanced questions about the evolution, function, and maintenance of phenotypic variation, for any trait that is repeatedly expressed.

KEYWORDS

brms, coefficient of variation, DHGLM, Double Hierarchical, location-scale regression, multivariate, repeatability, rstan

1 | INTRODUCTION

Life is full of variation. Phenotypic variation among taxa and species has been chronicled for centuries, but studying variation within populations, and even within individuals, is a newer venture for biologists (Westneat et al., 2015). While it is relatively straightforward to measure genetic differences between individuals, we cannot simply extrapolate from genetic variation to its phenotypic consequences (Frazer et al., 2009). Much phenotypic variation is

rooted in environmental variation (Stamps, 2015), either as adaptive responses to environmental change or maladaptive consequences of environmental stress (Snell-Rood, 2013), and individuals can differ in their responses (Dingemanse & Dochtermann, 2013). Even in benign environments phenotypes vary unpredictably (Hansen et al., 2006). For labile traits—which can be measured at multiple instances for the same individual—understanding what causes and maintains phenotypic variation both between and within individuals is a growing field (Mitchell et al., 2021).

Behavioural ecologists commonly use mixed models to measure how behaviours vary across environments, and between individuals within populations (Allegue et al., 2017). For non-human animals, behavioural traits that consistently vary between individuals have been deemed 'personality' traits, and sometimes these individual differences are correlated in 'behavioural syndromes' (e.g. some individuals are more risk-averse; Bell, 2007; Dingemanse et al., 2010; Dochtermann, 2010; Sih et al., 2004). Studies of individual differences in behaviour have generally revealed most behavioural variation is driven not by differences between individuals, but instead by residual variation (meta-analysis of repeatability ~ 0.37 ; Bell et al., 2009).

Standard mixed models assume homogeneity of residual variances. Residual variation represents both biological variability (e.g. within-individual variability) and measurement error. The homogeneity assumption is violated when some individuals are more variable than others across time (Ramakers et al., 2020). High 'heteroscedasticity' could represent measurement artefacts (e.g. individual differences in measurement error), non-adaptive deviations from an optimal phenotype (e.g. maladaptive imprecision; Hansen et al., 2006) or adaptive variation between individuals in their level of variability (e.g. alternative strategies; Wolf

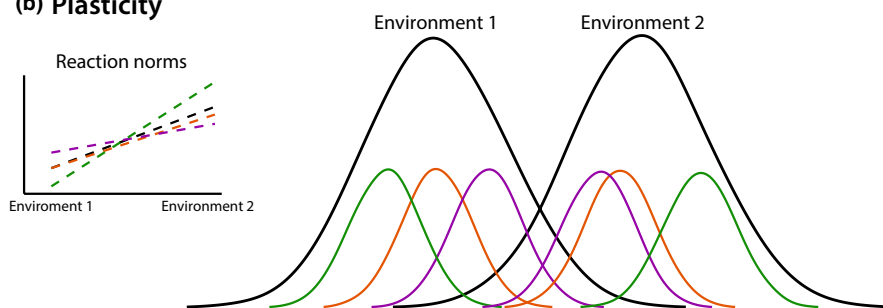
et al., 2007). We hereafter refer to an individual's level of variability in a given environment as 'predictability' (Cleasby et al., 2015). If biological mechanisms drive variation in predictability and are shared across different phenotypic traits, trade-offs could constrain predictability levels (e.g. individuals are more predictable than optimal for some traits, and less predictable than optimal for others; Pigliucci, 2003; Viney & Reece, 2013; Willmore et al., 2007).

Statistical methods for studying individual differences in labile (i.e. repeatedly expressed) traits will be most powerful when individual differences in averages (i.e. tendencies or personalities), plasticity, and predictability are considered together (Figure 1). Here, we provide a guide for empiricists on methods that can be used to study factors contributing to the evolution of phenotypic variation in labile traits, while lowering the barrier to entry with a reproducible worked example. Throughout this review, we describe models of behavioural traits (and therefore use terminology common in behavioural ecology), but the methods can be applied more broadly to different types of phenotypic traits, and different types of data clusters. For example, the clustering variable could be family or population origin rather than individual identity.

(a) Personality



(b) Plasticity



(c) Predictability

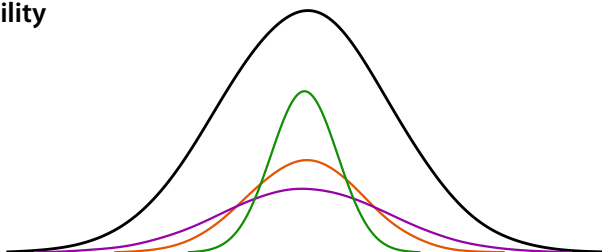


FIGURE 1 Conceptual illustration of three types of individual differences for a labile trait (in this case, behaviour). In each panel, black curves represent the normal distribution of a phenotypic trait in a population. Smaller, coloured curves represent the distribution of phenotypes expressed by an individual within that population. (a) 'Personality': individual differences in mean trait values, also known as phenotypic 'tendencies'. (b) 'Plasticity' due to a change in the environment (also known as 'flexibility' or 'responsiveness'). In environment 2, compared with environment 1, the average phenotype of the population increases, as shown by the black distribution shifting to the right. Individual differences in plasticity are shown by individual averages shifting to varying extents (i.e. variation in reaction norm slopes). (c) 'Predictability': individuals' level of variability (the breadth of individual distributions), also known as within- or intra-individual variability

2 | INDIVIDUAL DIFFERENCES IN PERSONALITY AND PLASTICITY

Personalities are usually quantified by including a random intercept for each individual in a mixed model. Other sources of variation can be modelled as fixed effects (and, if necessary, additional random effects). Throughout this paper, we will present Gaussian mixed models containing two fixed effects: the first for sex (i.e. a fixed effect with two categories, female and male) and a second for age (i.e. a continuous fixed effect). Age is mean-centred so that the overall intercept of the model represents the average phenotype of females at the average age of the population. Notations for all equations are explained in Table 1 (note that the same principles can be applied to non-Gaussian data too; Nakagawa & Schielzeth, 2010).

Non-human animal behaviours are commonly deemed 'personality traits' when, after measuring the same behaviour two or more times for multiple individuals, the differences among individuals are consistent across time and contexts (Bell, 2007; Sih et al., 2004). To measure differences in personalities, our basic model can be written as:

$$y_{ij} = (\beta_{m0} + ID_{m0j}) + \beta_{m1}x_{1j} + \beta_{m2}x_{2ij} + e_{ij}, \quad (1)$$

$$e_{ij} \sim (0, \sigma_e^2), \quad (2)$$

$$ID_{m0j} \sim (0, \sigma_{ID_{m0}}^2), \quad (3)$$

$$\sigma_{\text{fixed}_m}^2 = \text{var}(\beta_{m1}x_{1j} + \beta_{m2}x_{2ij}). \quad (4)$$

The model described by Equations 1–3 assumes homoscedasticity, meaning we model differences in personalities but not predictabilities (Figure 1a). The spread of personalities allows us to estimate the between-individual variance in behaviour, which is used to quantify the consistency of individual differences (equations for calculating repeatability and the coefficient of individual variation are provided in Section 4, below). When fixed effects represent biological variation (rather than experimental artefacts), it is recommended to add the fixed effect variance (calculated as in Equation 4) back into the total variance (de Villemereuil et al., 2018) before calculating repeatability.

When phenotypic traits are affected by an environmental or biological context (e.g. environmental temperature, hormone concentrations or biological age), we can model this relationship with a function called a 'reaction norm' (Gavrilets & Scheiner, 1993; Gomulkiewicz & Kirkpatrick, 1992; Stearns & Koella, 1986). In the simplest case of a linear relationship (specified by an intercept and slope), the slope (β_{m2}) describes the magnitude and direction of the population's average phenotypic plasticity. If the same individuals were measured multiple times across different contexts, we can use 'random regression' to estimate random slopes for each individual ($\beta_{m2} + ID_{m2j}$). Individuals can vary in both intercepts (personality; Figure 1a) and slopes (plasticity; Figure 1b). Consequently, the magnitude of differences in personality ($\sigma_{ID_{m0}}$) could depend upon the

context at which the intercept is estimated (in this case, the value of $x_2 = 0$, which is set to be the average age). In contrast to the model in Equation 1 (which assumed that individuals always maintain their ranking relative to the rest of the group), this 'random slope' model allows for individual rankings to change in different environments:

$$y_{ij} = (\beta_{m0} + ID_{m0j}) + \beta_{m1}x_{1j} + (\beta_{m2} + ID_{m2j})x_{2ij} + e_{ij}, \quad (5)$$

$$e_{ij} \sim (0, \sigma_e^2), \quad (6)$$

$$\begin{bmatrix} ID_{m0j} \\ ID_{m2j} \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{ID_{m0}}^2 & \rho(ID_{m0j}, ID_{m2j}) \sigma_{ID_{m0}} \sigma_{ID_{m2}} \\ \dots & \sigma_{ID_{m2}}^2 \end{bmatrix} \right). \quad (7)$$

Multiple individual differences are modelled together using the multivariate normal distribution (MVN), which estimates the covariance between the random intercepts and slopes across individuals (for simulations and discussion of what occurs when fitted data violate the MVN assumption, see Schielzeth et al., 2020). This covariance is written (in the upper triangle of Equation 7) as the product of the correlation between the intercepts and slopes [$\rho(ID_{m0j}, ID_{m2j})$], the standard deviation for the intercepts ($\sigma_{ID_{m0}}$) and standard deviation for the slopes ($\sigma_{ID_{m2}}$).

2.1 | Personality–plasticity associations

There are empirical observations of 'personality–plasticity associations', whereby individuals with different personalities differ in their plastic responses to environmental change. For example, in a marine gastropod, boldness was negatively correlated with plasticity in response to tidal and temperature changes (Cornwell et al., 2019); in sticklebacks, exploration was positively correlated with acclimation to a novel environment (Dingemanse et al., 2012); and in house sparrows, the level of parental care was shown to be correlated with plasticity in response to brood size, nestling age, precipitation and the provisioning effort of the breeding partner (Westneat et al., 2011).

Theoretically, Dubois (2019) predicted a negative correlation between proactive personalities and adaptive plasticity, based on the assumption that proactive individuals are less capable of accurately assessing their environment, due to the higher cognitive demands of proactivity. A positive correlation, meanwhile, could represent a 'rich get richer' scenario, whereby more well-resourced individuals are more proactive *and* better able to bear the costs associated with plasticity (DeWitt et al., 1998; Reznick et al., 2000). Alternatively, phenotypic plasticity can represent a maladaptive change in the phenotype (e.g. due to environmental stress), and therefore personality types that show reduced plasticity might be more resilient to environmental change (Ghalambor et al., 2007).

There are two possible types of personality–plasticity associations, the results of which are contrasted in Figure 2. First, from the MVN in Equation 7, we can ask whether individuals' personalities are correlated with individual differences in plasticity. The correlation provided by the

Notation	Definition
y_{ij}	Response variable (i.e. a behavioural trait): the measured phenotypic value of trait y for the j th individual at instance i
t_1	Superscript is used for bivariate models, to indicate model parameters for trait 1 (t_1) and trait 2 (t_2)
e_{ij}	Residual error: difference between the predicted and fitted value for the j th individual at instance i
σ_e^2	Residual variance for single hierarchical models ('mean' model only)
$\sigma_{e_i}^2$	Residual variance for double hierarchical models ('mean' and 'dispersion' models): unique value for each individual and instance
x_{1j}	Categorical input variable for the 'sex' of individual j ($x_{1j} = 0$ for female, and 1 for male)
x_{2ij}	Continuous input variable for the z-transformed 'age' of individual j at instance i ($x_{2ij} = 0$ is the average age of the population)
β_{m0}	Population intercept for the mean model. Average value of y when all other input variables are set to zero (females of average age)
$\beta_{v0,exp}$	Population intercept for the dispersion (variance) model. Average value of $\ln(\sigma_{e_{ij}}^2)$ when all other input variables are set to zero (females of average age). Estimated on the natural logarithm (ln) scale
β_{m1}	Population slope for the female–male contrast for the mean model
$\beta_{v1,exp}$	Population slope for the female–male contrast for the dispersion model. Estimated on the ln scale
β_{m2}	Population slope. Average value of phenotypic plasticity (reaction norm) for $x_{2ij} = z$ -scaled age, for the mean model
$\beta_{v2,exp}$	Population slope. Average value of phenotypic plasticity (reaction norm) for $x_{2ij} = z$ -scaled age, for the dispersion model. Estimated on the ln scale
ID_{m0j}	Difference between the population intercept β_{m0} and the random intercept for individual j for the mean model
$ID_{v0j,exp}$	Difference between the population intercept β_{v0} and the random intercept for individual j for the dispersion model. Estimated on the ln scale
ID_{m2j}	Difference between the population slope β_{m2} and the random slope for individual j for the mean model
$ \beta_{m2} + ID_{m2j} $	Absolute value of the (age) slope for individual j for the mean model. Describes the magnitude of individuals' average plasticity
$\sigma_{ID_{m0}}^2$	Between-individual variance for the individual intercepts for the mean model
$\sigma_{ID_{m2}}^2$	Between-individual variance for the individual slopes for the mean model
$\sigma_{ID_{v0,exp}}^2$	Between-individual variance for the individual intercepts for the dispersion model, on the ln scale
$\sigma_{fixed_m}^2$	Variance due to fixed effects for the mean model
$\sigma_{fixed_{v,exp}}^2$	Variance due to fixed effects for the dispersion model. Estimated on the ln scale
$\text{var}(a + b)$	Variance of the sum of random variables (vectors) a and b
$\rho(a, b)$	Correlation between two random variables a and b
σ_{ab}	Covariance between two random variables a and b

TABLE 1 Mathematical notation describing statistical models. Throughout this paper, we assume that we are modelling behavioural traits in a mixed model framework, and we are interested in the biological variables of sex, age and individual identity. Note that when presenting square matrices, the bottom triangle elements are omitted for simplicity (as they are identical to the upper triangle)

model is the ordinal association between individual differences (i.e. the best linear unbiased predictions: BLUPs) from the average population intercept (β_{m0}) and the average population slope (β_{m2}). This correlation represents the covariance between the random intercepts and slopes ($\sigma_{ID_{m0}ID_{m2}}$), divided by the product of their standard deviations:

$$\rho(ID_{m0j}, ID_{m2j}) = \frac{\sigma_{ID_{m0}ID_{m2}}}{\sigma_{ID_{m0}}\sigma_{ID_{m2}}}. \quad (8)$$

Alternatively, our question might be about the magnitude of plasticity irrespective of the direction of phenotypic change. For example, under thermal stress, are some individuals consistently better at maintaining homeostasis in physiological traits? The magnitude of plasticity is estimated as the absolute value of the summed population slope and individual slope difference, $|\beta_{m2} + ID_{m2j}|$. When fitting Bayesian mixed models, the correlation between the magnitude of

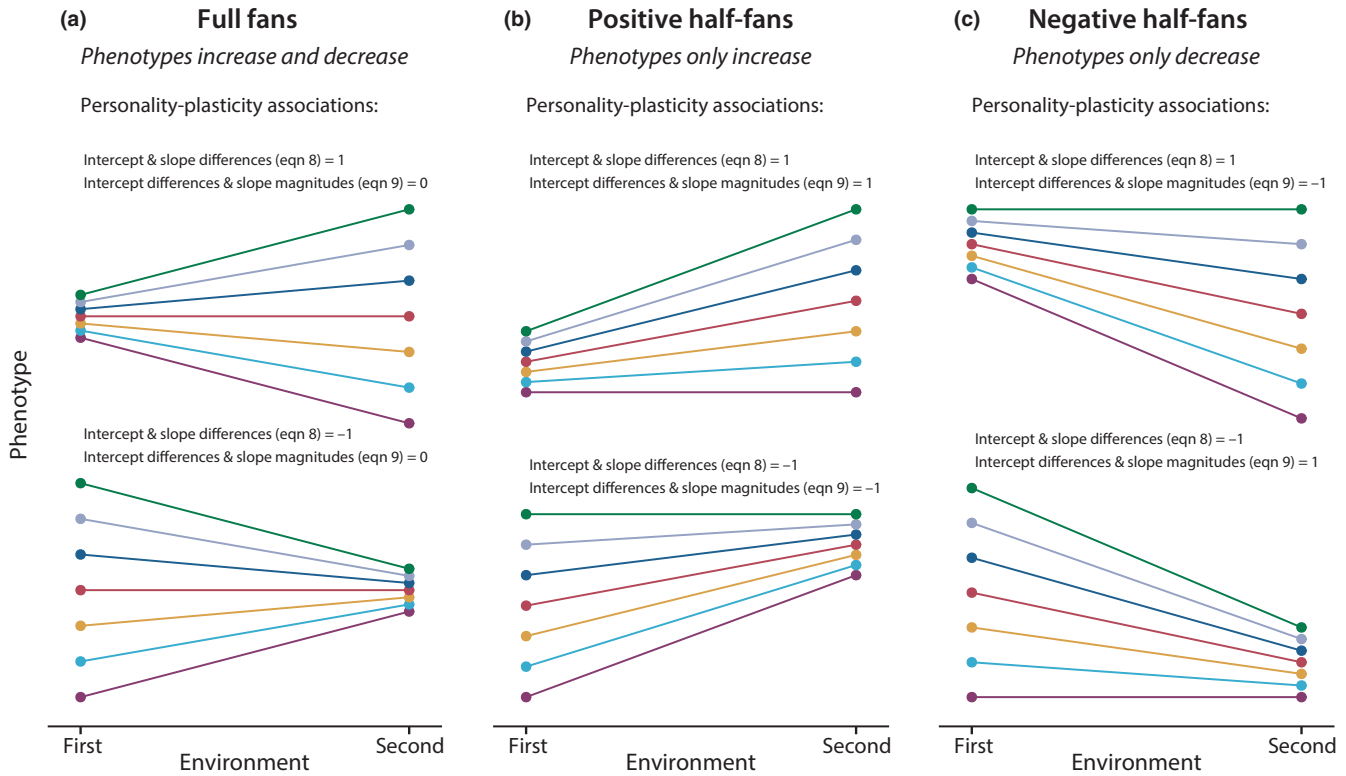


FIGURE 2 Personality-plasticity associations calculated with either slope differences, $\rho(\text{ID}_{m0j}, \text{ID}_{m2j})$ (Equation 8), or slope magnitudes, $\rho(\text{ID}_{m0j}, |\beta_{m2} + \text{ID}_{m2j}|)$ (Equation 9), for three simplified shapes of phenotypic plasticity. Associations are shown for a population of seven individuals, where phenotypes either ‘fan out’ (i.e. variance increases) or ‘fan in’ (i.e. variance decreases) across two environments. Points represent each individual’s average phenotype in two environments. Lines depict the direction and magnitude of phenotypic plasticity. Parameters estimated from models with random slopes are sensitive to the location we choose to set the intercept. While the ranking of individual averages is maintained across the two environments shown here, you can imagine extrapolating the lines into an environment where individuals who were below the population average are now above it, and vice versa. (a) Full fan: individuals vary in both the magnitude and direction of their slopes, meaning that some phenotypes increase in the second environment while others decrease. The personality–plasticity association is zero for slope magnitudes, positive for slope differences that fan out and negative for slope differences that fan in. (b) Positive fan: phenotypes always increase or stay the same in the second environment (i.e. individual slopes have a lower bound at zero). Personality–plasticity associations are identical for slope differences and magnitudes, with opposite signs for reaction norms that fan out or in (positive or negative correlations, respectively). (c) Negative fan: phenotypes always decrease or stay the same in the second environment (i.e. individual slopes have an upper bound at zero). Personality–plasticity associations are either positive or negative, depending both on whether slope differences or magnitudes are used, and whether the reaction norms fan in or out

each individual’s slope and the difference in their intercept from the population average,

$$\rho(\text{ID}_{m0j}, |\beta_{m2} + \text{ID}_{m2j}|) = \frac{\sigma_{\text{ID}_{m0j}|\beta_{m2} + \text{ID}_{m2j}|}}{\sigma_{\text{ID}_{m0j}} \sigma_{|\beta_{m2} + \text{ID}_{m2j}|}}, \quad (9)$$

can be calculated from the posterior distributions of individual differences, and the population slope. As for all calculations involving BLUPs, posterior distributions should be used when estimating Equation 9 to retain uncertainty and estimate credible intervals (Hadfield et al., 2010; Postma, 2006). While bootstrapping methods could be used to estimate uncertainty from frequentist (likelihood-based) models (cf. Stoffel et al., 2017), these methods would become very difficult when predictability is incorporated into the model structure.

Interpreting personality–plasticity associations at a given position of the intercept requires careful consideration, because multiple

patterns of reaction norm slopes can produce the same correlations (as shown in Figure 2, and noted by Stamps & Biro, 2016). A conceptual model of ‘fanning’ is described by Sih et al. (2015) as resulting from within-individual feedback loops. Fanning can also occur when adaptive plasticity is condition dependent, and only high-quality individuals can express adaptive plasticity. Individuals in poor condition (e.g. ill or injured) might express maladaptive plasticity in the opposite direction to the adaptive response. Regardless of the cause of these patterns, in a full fan scenario, the ranking of individual intercepts does not correlate with their magnitude of phenotypic plasticity (i.e. does not correlate with the absolute value of their slope). Contrasting with a full fan pattern, often we might expect all individuals in a population to respond to an environmental change with a plastic response in the *same* direction. In Figure 2, we call these scenarios ‘positive fans’ (when all phenotypes increase or stay the same) and ‘negative fans’ (when all phenotypes decrease or stay

the same). For example, ectotherms exposed to a warmer environment will often show a plastic response in the same direction (e.g. increased activity levels). Half-fans could be more likely to occur when the population average is close to a boundary (e.g. lower bound at zero), which is also likely to pose problems for the common assumption of residual normality.

2.2 | Bivariate model

When two different traits are measured repeatedly for the same individuals, we can use a bivariate model to estimate the covariances (and therefore correlations) between individual differences in personality and plasticity for these two traits (shown in Equation 13, below). Between-individual correlations that span across distinct traits might reflect integration preventing phenotypic traits from evolving independently (Fawcett et al., 2012; Pigliucci, 2003), such as genetic correlations (e.g. due to linkage disequilibrium) or developmental constraints (Sih et al., 2012). Trait correlations could also reflect correlated selective pressures, where a change in one trait encourages an adaptive change in the other. In theory, multivariate models can estimate the dependence between many traits at once. However, additional traits rapidly inflate the number of estimated covariances. Here—to reduce the computational and sample size burden, and for ease of presentation—we focus on the simplest scenario of two traits ('t1' and 't2'). The bivariate model can be written as:

$$y_{ij}^{t1} = \left(\beta_{m0}^{t1} + ID_{m0j}^{t1} \right) + \beta_{m1}^{t1} x_{1j}^{t1} + \left(\beta_{m2}^{t1} + ID_{m2j}^{t1} \right) x_{2j}^{t1} + e_{ij}^{t1}, \quad (10)$$

$$y_{ij}^{t2} = \left(\beta_{m0}^{t2} + ID_{m0j}^{t2} \right) + \beta_{m1}^{t2} x_{1j}^{t2} + \left(\beta_{m2}^{t2} + ID_{m2j}^{t2} \right) x_{2j}^{t2} + e_{ij}^{t2}, \quad (11)$$

$$\begin{bmatrix} e^{t1} \\ e^{t2} \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{e^{t1}}^2 & \rho(e^{t1}, e^{t2}) \sigma_{e^{t1}} \sigma_{e^{t2}} \\ \dots & \sigma_{e^{t2}}^2 \end{bmatrix} \right), \quad (12)$$

$$\begin{bmatrix} ID_{m0j}^{t1} \\ ID_{m2j}^{t1} \\ ID_{m0j}^{t2} \\ ID_{m2j}^{t2} \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{ID_{m0}^{t1}}^2 & \rho(ID_{m0j}^{t1}, ID_{m2j}^{t1}) \sigma_{ID_{m0}^{t1}} \sigma_{ID_{m2}^{t1}} & \rho(ID_{m0j}^{t1}, ID_{m0j}^{t2}) \sigma_{ID_{m0}^{t1}} \sigma_{ID_{m0}^{t2}} & \rho(ID_{m0j}^{t1}, ID_{m2j}^{t2}) \sigma_{ID_{m0}^{t1}} \sigma_{ID_{m2}^{t2}} \\ \dots & \sigma_{ID_{m2}^{t1}}^2 & \rho(ID_{m2j}^{t1}, ID_{m0j}^{t2}) \sigma_{ID_{m2}^{t1}} \sigma_{ID_{m0}^{t2}} & \rho(ID_{m2j}^{t1}, ID_{m2j}^{t2}) \sigma_{ID_{m2}^{t1}} \sigma_{ID_{m2}^{t2}} \\ \dots & \dots & \sigma_{ID_{m0}^{t2}}^2 & \rho(ID_{m0j}^{t2}, ID_{m2j}^{t2}) \sigma_{ID_{m0}^{t2}} \sigma_{ID_{m2}^{t2}} \\ \dots & \dots & \dots & \sigma_{ID_{m2}^{t2}}^2 \end{bmatrix} \right). \quad (13)$$

Dependence between residual errors for different traits is modelled using the MVN in Equation 12. Similarly, in Equation 13, the covariance matrix describing the relationship between individual-level differences has been expanded to include correlations both within and between traits.

2.3 | Between-trait correlation: Behavioural syndromes

Bivariate models quantify relationships between two traits (Equations 10–13). When personality traits are correlated, they are said to exhibit a 'behavioural syndrome' (Dingemanse et al., 2010), which we can estimate as:

$$\rho(ID_{m0j}^{t1}, ID_{m0j}^{t2}) = \frac{\sigma_{ID_{m0}^{t1} ID_{m0}^{t2}}}{\sigma_{ID_{m0}^{t1}} \sigma_{ID_{m0}^{t2}}}. \quad (14)$$

While many empirical papers purport to have found these syndromes, far fewer have done so following the recommended method of decomposing total phenotypic variance into its between- and within-individual components (Dingemanse & Dochtermann, 2013; Moirón et al., 2020; Niemelä & Dingemanse, 2018). Combining both levels of the phenotypic correlation can be misleading because their strength and direction can differ (i.e. violating the 'individual gambit'; Brommer, 2013). Whereas both between- and within-individual correlations can be caused by environmental effects, only between-individual correlations can harbour additive genetic covariances.

2.4 | Between-trait correlation: Plasticity syndromes

Between-individual plasticity correlations can be measured for multiple traits, or multiple environmental manipulations. Positive correlations could be caused by shared mechanisms in the maintenance of plasticity; the plant sciences have long studied plasticity integration (Gianoli & Palacio-Lopez, 2009; Mallitt et al., 2010; Pigliucci, 2002; Schlichting, 1989). Alternatively, a negative correlation in the magnitude of plasticity could reflect trade-offs due to associated costs (DeWitt et al., 1998), while the absence of a correlation suggests the traits are decoupled (e.g. face-independent selective pressures).

'Plasticity syndromes' are more challenging to interpret than behavioural syndromes, due to the rankings of individual differences in

slopes not necessarily corresponding with the magnitude of individuals' plasticity. As with personality–plasticity associations, plasticity syndromes can be estimated in two different ways (which are compared in Figure S1, Supporting Information). Taken directly from the model, the correlation between individual slope differences,

$$\rho\left(\text{ID}_{m2j}^{t1}, \text{ID}_{m2j}^{t2}\right) = \frac{\sigma_{\text{ID}_{m2}^{t1}\text{ID}_{m2}^{t2}}}{\sigma_{\text{ID}_{m2}^{t1}} \sigma_{\text{ID}_{m2}^{t2}}}, \quad (15)$$

describes whether the order of slopes is maintained between the two traits. When Equation 15 is positive, individuals whose slopes are more positive than average in trait 1 tend to *also* be more positive than average in trait 2. Quantifying the maintenance of rankings is useful for certain patterns of plasticity. For example, imagine in response to a low-quality diet the activity of some digestive enzymes decreases (negative slopes for trait 1, negative half-fan). Some individuals will be able to compensate with increased foraging effort (trait 2) and show less change in enzyme activity, while those in poor condition might show reduced foraging effort as they conserve energy alongside a greater decrease in enzyme activity (i.e. both negative and positive slopes for trait 2, resulting in a positive correlation from Equation 15).

We can imagine other scenarios where slope steepness is of greater interest than individual differences from the average slope (e.g. maintaining homeostasis for multiple traits under thermal stress). In this case, a 'plasticity syndrome' (Equation 16) is calculated as the correlation between the absolute magnitude of individuals' reaction norms, such that:

$$\rho\left(\left|\beta_{m2}^{t1} + \text{ID}_{m2j}^{t1}\right|, \left|\beta_{m2}^{t2} + \text{ID}_{m2j}^{t2}\right|\right) = \frac{\sigma_{\left|\beta_{m2}^{t1} + \text{ID}_{m2j}^{t1}\right| \left|\beta_{m2}^{t2} + \text{ID}_{m2j}^{t2}\right|}}{\sigma_{\left|\beta_{m2}^{t1} + \text{ID}_{m2j}^{t1}\right|} \sigma_{\left|\beta_{m2}^{t2} + \text{ID}_{m2j}^{t2}\right|}}. \quad (16)$$

As with Equation 9, correlations involving absolute values of slopes can be calculated from the posterior distributions of model estimates.

2.5 | Summary of personality and plasticity

Individual differences in personality and plasticity produce three types of biologically relevant correlations: first, personality-plasticity associations are a correlation between reaction norm intercepts and slope differences or magnitudes; second, behavioural syndromes are a correlation between individual intercepts for more than one trait; third, plasticity syndromes are a correlation between slope differences or magnitudes for more than one trait, or the same trait measured across more than one covariate. Individual differences in plasticity can cause estimates of personality and related correlations to differ, depending on the biological interpretation of the intercept. When interpreting ordinal associations involving slopes, which have both a direction and magnitude, researchers should plot each individual's reaction norm to consider the 'shape' of

phenotypic plasticity. For some research questions, the magnitude of plasticity could be more relevant than the direction of change away from the population average. In these circumstances, researchers can perform additional calculations to capture the absolute value of individual slopes, rather than individual differences from the average slope. Performing vector calculations on posterior distributions (from a Bayesian model) ensures that uncertainty in model estimates is carried forward.

3 | INDIVIDUAL DIFFERENCES IN PREDICTABILITY

The effect animals have on their surroundings depends not only on their average behaviour, but also on how their behaviour fluctuates through time. Individual differences can be consistent yet small, and these might not have a material impact on fitness (and therefore might not respond to selection). Despite the variability of individuals' behaviour being biologically important, it is currently rare for behavioural studies to distinguish between individuals who are very consistent through time, and those whose behaviour fluctuates enormously (an early example is seen in Westneat et al., 2013). Individual differences in predictability can be modelled with a Double Hierarchical Generalised Linear Model (DHGLM; Cleasby et al., 2015). The 'double' in DHGLM refers to a random effect being included in both the mean and dispersion models. The dispersion model—also known as the residual variance model—is usually estimated on the natural logarithm scale. In the social and medical sciences, DHGLMs are also known as location-scale regression models (with 'location' indicating the mean, and 'scale' indicating the variance; e.g. Lin et al., 2018; Rast et al., 2012). Fitting a random intercept for individual identity at both levels of the model allows individuals to vary in both personality (Figure 1a) and predictability (Figure 1c).

3.1 | Modelling individual distributions

Extending the univariate model shown in Equations 5–8, we can write the double hierarchical model as:

$$y_{ij} = (\beta_{m0} + \text{ID}_{m0j}) + \beta_{m1}x_{1j} + (\beta_{m2} + \text{ID}_{m2j})x_{2j} + e_{ij}, \quad (17)$$

$$\ln(\sigma_{e_{ij}}^2) = (\beta_{v0,\text{exp}} + \text{ID}_{v0j,\text{exp}}) + \beta_{v1,\text{exp}}x_{1j} + \beta_{v2,\text{exp}}x_{2j}, \quad (18)$$

$$e_{ij} \sim N(0, \sigma_{e_{ij}}^2), \quad (19)$$

$$\begin{bmatrix} \text{ID}_{m0j} \\ \text{ID}_{v0j,\text{exp}} \\ \text{ID}_{m2j} \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{\text{ID}_{m0}}^2 & \rho(\text{ID}_{m0j}, \text{ID}_{v0j,\text{exp}}) \sigma_{\text{ID}_{m0}} \sigma_{\text{ID}_{v0,\text{exp}}} & \rho(\text{ID}_{m0j}, \text{ID}_{m2j}) \sigma_{\text{ID}_{m0}} \sigma_{\text{ID}_{m2}} \\ \dots & \sigma_{\text{ID}_{v0,\text{exp}}}^2 & \rho(\text{ID}_{v0j}, \text{ID}_{m2j}) \sigma_{\text{ID}_{v0,\text{exp}}} \sigma_{\text{ID}_{m2}} \\ \dots & \dots & \sigma_{\text{ID}_{m2}}^2 \end{bmatrix} \right). \quad (20)$$

Estimating individual variances requires many repeated measurements at the individual level, which is relatively uncommon in animal personality studies (sample size recommendations depend on the number of individuals and the magnitude of heteroscedasticity, which is explored in Cleasby et al., 2015). Note that Equations 17–20 vary from Equations 19–24 in Cleasby et al. (2015), as the dispersion model is based on residual variances, rather than residual standard deviations (which has some benefits for summarising the magnitude of individual differences; see Section 4.3, below).

3.2 | Within-trait correlations between personality, plasticity and predictability

From the correlation between individual intercepts in both the mean and dispersion models, we can estimate whether some personality types are more prone to being unpredictable than others. From the multivariate distribution in Equation 20, we have:

$$\rho(ID_{m0j}, ID_{v0j,exp}) = \frac{\sigma_{ID_{m0}ID_{v0,exp}}}{\sigma_{ID_{m0}}\sigma_{ID_{v0,exp}}}. \quad (21)$$

Interpreting Equation 21 is somewhat unintuitive; remember that an individual having more residual variance is less predictable. Therefore, a positive correlation between mean and dispersion intercepts represents a negative correlation between personality and predictability. When presenting results, we prefer to multiply correlations involving dispersion intercepts by -1 , to make their interpretation intuitive (e.g. a positive correlation signifies a bolder individual is more predictable, with a smaller residual variance), such that:

$$\rho(ID_{m0j}, -ID_{v0j,exp}) = -\frac{\sigma_{ID_{m0}ID_{v0,exp}}}{\sigma_{ID_{m0}}\sigma_{ID_{v0,exp}}}. \quad (22)$$

Our supplementary example presents this sign-reversed correlation for personality–predictability associations. Although little theory exists on the personality–predictability association, we might expect riskier personality types to be less predictable (as being more variable can be a risky strategy). Alternatively, riskier individuals could be closer to a hypothetical ‘ceiling’, whereby a fluctuation beyond that point would be fatal to the individual. Riskier individuals might therefore show greater precision around their mean phenotype, to avoid crossing some point of no return (a similar idea around stability of more ‘extreme’ personalities is discussed in Stamps & Groothuis, 2010).

Broadly, plasticity is the expression of different phenotypes by the same genotype in a different environment (Stamps, 2015). The environment will always be slightly different each time an individual expresses a labile trait because of variation in endogenous variables (internal and developmental), and uncontrolled fluctuations in the external environment (Flatt, 2005; Hansen et al., 2006). Therefore, predictability is a special type of ‘stochastic plasticity’; there are stochastic changes in

internal and external environments that prevent us from knowing exactly which phenotype will be expressed at any point in time. From the slope in the mean model and the intercept in the dispersion model, we can estimate whether individual differences in traditional and stochastic plasticity are correlated. There is theoretical interest in whether different types of plasticity are related to each other but to date this type of question has received little empirical attention (Stamps & Biro, 2016). For a given trait and a given environment, less predictable individuals have a wider range of trait expressions. This range could be correlated with a stronger plastic response when exposed to a different environment. The correlation between ordered individual differences from mean slopes and dispersion intercepts,

$$\rho(ID_{m2j}, ID_{v0j,exp}) = \frac{\sigma_{ID_{m2}ID_{v0,exp}}}{\sigma_{ID_{m2}}\sigma_{ID_{v0,exp}}}, \quad (23)$$

measures whether individuals that are further away from the average level of plasticity are more or less predictable than average. The correlation between the magnitudes of mean slopes and dispersion intercepts,

$$\rho(|\beta_{m2} + ID_{m2j}|, -ID_{v0j,exp}) = -\frac{\sigma_{|\beta_{m2} + ID_{m2}|ID_{v0,exp}}}{\sigma_{|\beta_{m2} + ID_{m2}|}\sigma_{ID_{v0,exp}}}, \quad (24)$$

estimates whether individuals who are more plastic (in either direction) are more or less predictable. The minus term makes this correlation interpretable as a ‘plasticity–predictability association’.

3.3 | Between-trait correlation: Predictability syndromes

Up to this point, we have discussed five types of correlations between individual differences: behavioural syndromes (Figure 3a), plasticity syndromes (Figure 3b), personality–plasticity associations (Figure 3d), personality–predictability associations (Figure 3e) and plasticity–predictability associations (Figure 3f). Given sufficient data, a sixth correlation can be estimated simultaneously: predictability syndromes (Figure 3c). The bivariate model can be written as:

$$y_{ij}^{t1} = (\beta_{m0}^{t1} + ID_{m0j}^{t1}) + \beta_{m1}^{t1}x_{1j}^{t1} + (\beta_{m2}^{t1} + ID_{m2j}^{t1})x_{2j}^{t1} + e_{ij}^{t1}, \quad (25)$$

$$y_{ij}^{t2} = (\beta_{m0}^{t2} + ID_{m0j}^{t2}) + \beta_{m1}^{t2}x_{1j}^{t2} + (\beta_{m2}^{t2} + ID_{m2j}^{t2})x_{2j}^{t2} + e_{ij}^{t2}, \quad (26)$$

$$\ln\left(\sigma_{e_{ij}^{t1}}^2\right) = (\beta_{v0}^{t1} + ID_{v0j,exp}^{t1}) + \beta_{v1}^{t1}x_{1j}^{t1} + \beta_{v2}^{t1}x_{2j}^{t1}, \quad (27)$$

$$\ln\left(\sigma_{e_{ij}^{t2}}^2\right) = (\beta_{v0}^{t2} + ID_{v0j,exp}^{t2}) + \beta_{v1}^{t2}x_{1j}^{t2} + \beta_{v2}^{t2}x_{2j}^{t2}, \quad (28)$$

$$\begin{bmatrix} e_{ij}^{t1} \\ e_{ij}^{t2} \end{bmatrix} \sim \text{MVN} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{e_{ij}^{t1}}^2 & \rho(e_{ij}^{t1}, e_{ij}^{t2})\sigma_{e_{ij}^{t1}}\sigma_{e_{ij}^{t2}} \\ \dots & \sigma_{e_{ij}^{t2}}^2 \end{bmatrix} \right), \quad (29)$$

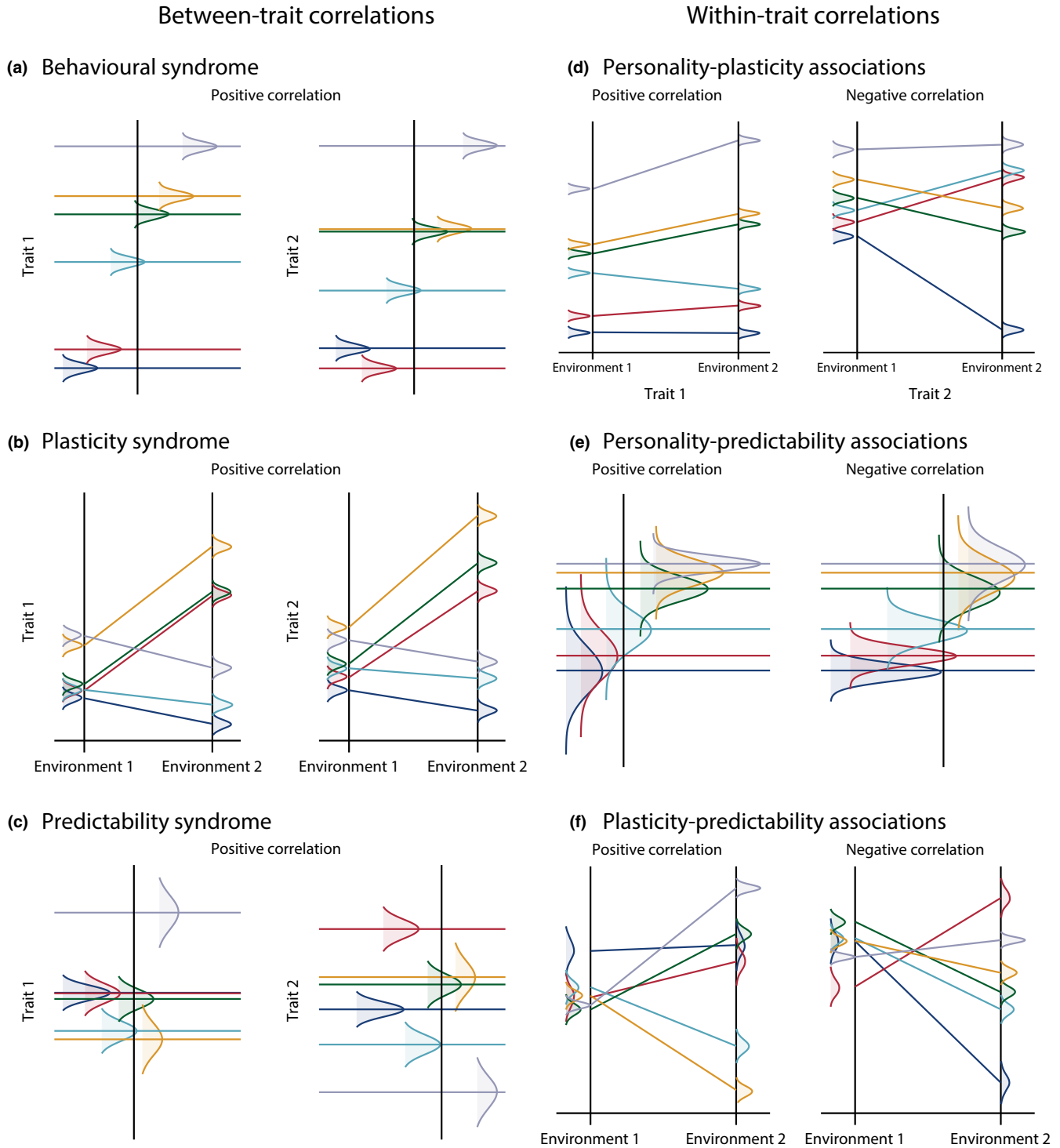


FIGURE 3 Conceptual illustration of six types of correlations, from individual differences in personality, plasticity and predictability. Each coloured line and distribution represents a different individual from the same population. The left column (panels a–c) shows positive between-trait correlations ('syndromes'), where individual differences are correlated with each other for multiple traits. The right column (panels d–f) shows within-trait correlations between pairs of individual differences. (a) Behavioural syndrome: individual differences in personality (measured by random intercepts) are positively correlated between two traits, meaning that the 'rank order' of intercepts is maintained (Equation 14). (b) Plasticity syndrome: the magnitudes of random slopes are positively correlated (Equation 16). (c) Predictability syndrome: individuals that are less predictable in one trait (shown by a wider distribution) are less predictable in the second trait (Equation 31). (d) Personality–plasticity association: individuals with a higher personality ranking (more positive intercept) have larger *absolute* slopes (Equation 9). (e) Personality–predictability association: individuals' personality (intercepts) are correlated with their level of predictability (their reversed magnitude of within-individual variance; Equation 22). (f) Plasticity–predictability syndrome: the magnitude of individual slopes correlates with the ranking of predictability (reversed within-individual variance; Equation 24)

$$\begin{bmatrix} ID_{m0j}^{t1} \\ ID_{v0j,exp}^{t1} \\ ID_{m2j}^{t1} \\ ID_{m0j}^{t2} \\ ID_{v0j,exp}^{t2} \\ ID_{m2j}^{t2} \end{bmatrix} \sim MVN \left(\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{ID_{m0j}^{t1}}^2 & \rho(ID_{m0j}^{t1}, ID_{v0j,exp}^{t1}) \sigma_{ID_{m0j}^{t1}} \sigma_{ID_{v0j,exp}^{t1}} & \rho(ID_{m0j}^{t1}, ID_{m2j}^{t1}) \sigma_{ID_{m0j}^{t1}} \sigma_{ID_{m2j}^{t1}} & \rho(ID_{m0j}^{t1}, ID_{v0j,exp}^{t2}) \sigma_{ID_{m0j}^{t1}} \sigma_{ID_{v0j,exp}^{t2}} & \rho(ID_{m0j}^{t1}, ID_{m2j}^{t2}) \sigma_{ID_{m0j}^{t1}} \sigma_{ID_{m2j}^{t2}} \\ \dots & \sigma_{ID_{v0j,exp}^{t1}}^2 & \rho(ID_{v0j,exp}^{t1}, ID_{m2j}^{t1}) \sigma_{ID_{v0j,exp}^{t1}} \sigma_{ID_{m2j}^{t1}} & \rho(ID_{v0j,exp}^{t1}, ID_{v0j,exp}^{t2}) \sigma_{ID_{v0j,exp}^{t1}} \sigma_{ID_{v0j,exp}^{t2}} & \rho(ID_{v0j,exp}^{t1}, ID_{m2j}^{t2}) \sigma_{ID_{v0j,exp}^{t1}} \sigma_{ID_{m2j}^{t2}} \\ \dots & \dots & \sigma_{ID_{m2j}^{t1}}^2 & \rho(ID_{m2j}^{t1}, ID_{v0j,exp}^{t2}) \sigma_{ID_{m2j}^{t1}} \sigma_{ID_{v0j,exp}^{t2}} & \rho(ID_{m2j}^{t1}, ID_{m2j}^{t2}) \sigma_{ID_{m2j}^{t1}} \sigma_{ID_{m2j}^{t2}} \\ \dots & \dots & \dots & \sigma_{ID_{m0j}^{t2}}^2 & \rho(ID_{m0j}^{t2}, ID_{v0j,exp}^{t2}) \sigma_{ID_{m0j}^{t2}} \sigma_{ID_{v0j,exp}^{t2}} & \rho(ID_{m0j}^{t2}, ID_{m2j}^{t2}) \sigma_{ID_{m0j}^{t2}} \sigma_{ID_{m2j}^{t2}} \\ \dots & \dots & \dots & \dots & \sigma_{ID_{v0j,exp}^{t2}}^2 & \rho(ID_{v0j,exp}^{t2}, ID_{m2j}^{t2}) \sigma_{ID_{v0j,exp}^{t2}} \sigma_{ID_{m2j}^{t2}} \\ \dots & \dots & \dots & \dots & \dots & \sigma_{ID_{m2j}^{t2}}^2 \end{bmatrix} \right) \quad (30)$$

The variance–covariance matrix in Equation 30 emphasises, in bold, the off-diagonal elements that comprise the six types of correlations we are interested in (shown in Figure 3).

Predictability syndromes describe whether individuals' level of predictability in one trait correlates with their level of predictability in a second trait, such that:

$$\rho(ID_{v0j,exp}^{t1}, ID_{v0j,exp}^{t2}) = \frac{\sigma_{ID_{v0j,exp}^{t1} ID_{v0j,exp}^{t2}}}{\sigma_{ID_{v0j,exp}^{t1}} \sigma_{ID_{v0j,exp}^{t2}}}. \quad (31)$$

(Following the notations described in Table 1, the numerator $\sigma_{ID_{v0j,exp}^{t1} ID_{v0j,exp}^{t2}}$ is the covariance between $ID_{v0j,exp}^{t1}$ and $ID_{v0j,exp}^{t2}$, while the denominator $\sigma_{ID_{v0j,exp}^{t1}} \sigma_{ID_{v0j,exp}^{t2}}$ is the product of their standard deviations). The presence of a 'predictability syndrome' could imply integration (which might represent correlated selective pressures or genetic correlations; Pigliucci, 2003), or correlations could be an artefact of measurement error (e.g. the labile traits of smaller or more active individuals might be recorded with lower precision). The absence of a predictability syndrome implies that different types of traits might be selected to have different levels of predictability.

3.4 | Introducing stochastic malleability

As a future extension to the methods reviewed here, it is possible (given sufficient data) to include a random slope in the dispersion model (i.e. to add $ID_{v2j,exp}$ into Equation 18), to estimate individual differences in 'stochastic malleability' (i.e. plasticity in predictability, or simply 'malleability'). While it would require many repeated measurements across different contexts (data simulations are required to estimate the minimum sample size requirements), a fourth type of individual difference, in malleability, could answer three additional questions (Figure 4, below): (a) is the level of malleability correlated across traits (i.e. *malleability syndromes*), or can individuals be malleable in one trait and show fixed predictability in another? (b): do individuals with more plasticity in personality show more plasticity in variability (i.e. *plasticity-malleability associations*)? (c) are some personality types more or less likely to change their level of predictability in response to an environmental change (i.e. *personality-malleability associations*)? Stochastic malleability could be an important aspect of learning or adapting to novel conditions: naïve individuals (i.e. individuals who are young, or in an unfamiliar environment) might increase variability to 'sample' a wider array of options. As individuals gain more experience, they might hone in upon the optimal phenotype, and therefore become more

predictable (McNamara et al., 2006). An interesting avenue of future research, therefore, could be to incorporate individual differences in malleability into studies of learning or invasion biology (c.f. Chapple et al., 2012; Griffin et al., 2015).

3.5 | Summary of predictability

With two individual differences –a random intercept and slope in the mean model to quantify personality and plasticity—we can look at three correlations: two types of syndromes (between traits; Figure 3a,b) and one intercept-slope association (within trait; Figure 3d). Modelling predictability adds a third individual difference –a random intercept in the dispersion model. Using a bivariate (multivariate) model, we can simultaneously model these three individual differences in two (or more) types of traits (Equations 25–30), and estimate three additional correlations: (a) a predictability syndrome (between traits; Figure 3c); (b) an association (within traits) between personality and predictability (Figure 3e); and (c) an association between plasticity and predictability (Figure 3f). With adequate sampling designs and statistical power, this model can be extended to quantify how much individuals differ in their change in predictability in different contexts (i.e. 'stochastic malleability'; Figure 4).

4 | SUMMARY STATISTICS FOR META-ANALYSIS

The preceding sections described how mixed models can be used to quantify individual differences in personality, plasticity, and predictability, but how can we compare our results to those from other studies? For between-study comparisons and synthesis (including meta-analyses), the magnitude of individual differences in personality and predictability can be quantified with two different summary statistics: repeatability (R_p), which is variance-standardised, and the coefficient of individual variation (CV_{ID}), which is mean-standardised. The coefficient of individual variation is suitable for ratio-scale measurements (i.e. variables with a true zero and equal intervals between neighbours points, such as number of offspring or total activity time; Houle et al., 2011), although Hansen et al. (2011) discuss how mean-standardisation can also be done with log-interval and signed-interval scales.

For ratio-scale data, both repeatability and the coefficient of individual variation are phenotypic analogues for statistics relating

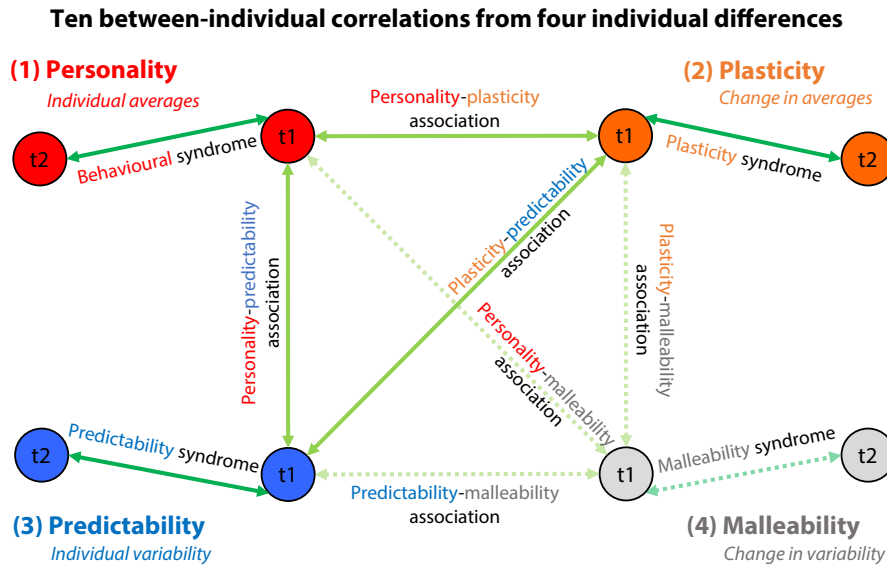


FIGURE 4 Ten types of interpretable between-individual correlations can be modelled in a bivariate DHGLM (t_1 = trait 1 and t_2 = trait 2; as in Table 1), containing four individual differences: (1) personality (random intercept in mean models); (2) plasticity (random slope in mean models); (3) predictability (random intercept in dispersion models); and (4) malleability (random slope in dispersion models). Solid lines indicate correlations that were modelled in the supplementary worked example; our dataset was not suitable to model the correlations shown by dashed lines. Note that a covariance matrix for eight individual differences would estimate 28 correlations total (18 more than those named here, which are hard to interpret). Any correlation from such a model should be interpreted cautiously, given multiple comparisons inflate the rate of spurious associations

to evolutionary potential (Houle, 1992). Repeatability roughly sets the upper limit on narrow-sense heritability (but see Dohm, 2002), whereas the coefficient of individual variation is analogous to the coefficient of additive genetic variance, CV_A (Dochtermann & Royauté, 2019; Holtmann et al., 2017; Houle, 1992). A repeatability estimate from the dispersion model, Rp_v , will always be smaller than its counterpart from the mean model, Rp_m (because the denominator for Rp_v includes a term that multiplies the numerator by more than three, which is not the case for Rp_m ; see the equations below), whereas estimates of the coefficient of individual variation for means and variances are more comparable to each other. We recognise that the utility of these evolutionary potential statistics is debatable (e.g. Hansen et al. (2011) argued that the square of the coefficient of variation, I_A , has a more general interpretation and is therefore more relevant than CV_A).

Below, we describe the calculations required to obtain Rp and CV_{ID} from DHGLM model described by Equations 17–20. Supplementary R code (O'Dea et al., 2021) is available to calculate Rp and CV_{ID} for all models described above and, with some minor modifications, the formulas are broadly applicable for other model specifications too.

4.1 | Repeatability and the coefficient of individual variation

Repeatability for the mean model (Rp_m) and dispersion model (Rp_v) are given by:

$$Rp_m = \frac{\sigma_{ID_m}^2}{\sigma_p^2}, \quad (32)$$

$$Rp_v = \frac{\sigma_{ID_v}^2}{\sigma_p^2}, \quad (33)$$

where σ_p^2 is the total phenotypic variance, σ_p^2 is the total variance in phenotypic variance, and $\sigma_{ID_m}^2$ and $\sigma_{ID_v}^2$ are the variance components for between-individual differences in the mean and dispersion models, respectively (Nakagawa & Schielzeth, 2010).

Coefficients of individual variation (similar to CV for additive genetic variance; Mulder et al., 2007; Sae-Lim et al., 2015) for the mean model (CV_{ID_m}) and dispersion model (CV_{ID_v}) are given by:

$$CV_{ID_m} = \frac{\sigma_{ID_{\mu_p}}}{\mu_p}, \quad (34)$$

$$CV_{ID_v} = \frac{\sigma_{ID_v}}{\bar{\sigma}_w^2}. \quad (35)$$

where μ_p is the average individual phenotype, $\bar{\sigma}_w^2$ is the average within-individual variance (the 'w' represents 'within', and the bar represents the average), and $\sigma_{ID_{\mu_p}}$ and σ_{ID_v} are the standard deviations for between-individual differences in the mean and dispersion models, respectively. If no transformations have been applied to the response variable, y , then $\sigma_{ID_{\mu_p}} = \sqrt{\sigma_{ID_m}^2}$ (i.e. the square root of the numerator for repeatability of the mean, Equation 32), and the population mean is calculated for an even sex ratio at the average age of the population ($\mu_p = \frac{2\beta_{m0} + \beta_{m1}}{2}$).

4.2 | Obtaining each parameter

4.2.1 | Converting parameters from the dispersion model

When calculating R_p and CV_{ID} from DHGLM models, it is essential that all parameters from the dispersion model are first converted back from the natural logarithm (ln) scale onto the same scale as the mean model so that variance terms can be summed. In general, if we have a mean and variance that are estimated on the ln scale, $\mu_{y,exp}$ and $\sigma_{y,exp}^2$, then we can convert them back to the normal (observed) scale as follows:

$$\mu_y = \exp\left(\mu_{y,exp} + \frac{\sigma_{y,exp}^2}{2}\right), \quad (36)$$

$$\sigma_y^2 = \left(\exp\left(\sigma_{y,exp}^2\right) - 1\right) \exp\left(2\mu_{y,exp} + \sigma_{y,exp}^2\right), \quad (37)$$

where μ_y and σ_y^2 are the mean and variance on the observed scale. Note that simply taking the exponent of the mean on the ln scale, $\exp(\mu_{y,exp})$, gives the median estimate on the observed scale, rather than the mean.

4.2.2 | Within-individual variance

Usually, the within-individual variance $\bar{\sigma}_w^2$ is assumed to be equal to the average residual variance, $\bar{\sigma}_e^2$. However, there could be a scenario where we calculate $\bar{\sigma}_w^2 < \bar{\sigma}_e^2$ by removing an artificial source of variance from the dispersion model (e.g. estimated measurement error). For now, let us assume all the variance in y is biologically meaningful (i.e. we assume $\sigma_p^2 = \sigma_y^2$) (de Villemereuil et al., 2018). We therefore take the total variance from the dispersion model as $\sigma_{v,exp}^2 = \sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2$.

On the ln-normal scale, the mean residual variance is the 'population intercept' from the dispersion model, $\beta_{pv0,exp} = \frac{2\beta_{v0,exp} + \beta_{v1,exp}}{2}$, assuming an equal sex ratio with individuals at an average age, $x_{2ij} = 0$ (where $\beta_{v0,exp}$ is the female intercept, and $\beta_{v1,exp}$ is the female-male contrast; Table 1). By substituting the ln-normal mean and variance into the mean conversion formula for a ln-normal distribution (i.e. μ_y in Equation 36), we obtain $\bar{\sigma}_w^2$ as:

$$\bar{\sigma}_w^2 = \exp\left(\beta_{pv0,exp} + \frac{\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2}{2}\right). \quad (38)$$

Different model structures will require modifications of the above (and below) equations, for example, when $\sigma_y^2 \neq \sigma_p^2$ and/or $\bar{\sigma}_e^2 \neq \bar{\sigma}_w^2$.

4.2.3 | Between-individual variance and total phenotypic variance

The variance components from the mean model (including variance due to fixed effects) can be summed to obtain $\sigma_{ID_m}^2$ and σ_p^2 (Allegue

et al., 2017). In our case (Equations 17–20), modelling individual differences in intercepts (ID_{m0}) and slopes (ID_{m2}) across age (x_2), the variances are written as:

$$\sigma_{ID_m}^2 = \sigma_{ID_{m0}}^2 + \sigma_{ID_{m2}}^2 \sigma_{x_2}^2 + \sigma_{ID_{m2}}^2 \mu_{x_2}^2 + \rho(ID_{m0}, ID_{m2}) \sigma_{ID_{m0}} \sigma_{ID_{m2}} 2\mu_{x_2}, \quad (39)$$

$$\sigma_p^2 = \sigma_{ID_m}^2 + \sigma_{fixed_m}^2 + \bar{\sigma}_w^2, \quad (40)$$

$$x_{2ij} \sim D(\mu_{x_2}, \sigma_{x_2}^2). \quad (41)$$

The predictor variable x_2 has a mean of μ_{x_2} and a variance of $\sigma_{x_2}^2$ with an arbitrary distribution, D (because no assumptions are made about the distribution of predictors). From Equation 39, we can see that when individual differences in personality and plasticity are modelled at the same time, the magnitude of individual differences will depend upon the 'environment' or 'context' at which intercepts are estimated. Typically, continuous predictor variables are mean-centred so that intercepts are estimated at the average value for that trait ($\mu_{x_2} = 0$). When the predictor is also z-transformed ($\sigma_{x_2}^2 = 1$), the between-individual variance is simply $\sigma_{ID_m}^2 = \sigma_{ID_{m0}}^2 + \sigma_{ID_{m2}}^2$ (this is the case in our worked example; Supporting Information).

4.2.4 | Variance in total phenotypic variance

To calculate variance of the total phenotypic variance, $\sigma_{\sigma_p^2}^2$, we first need to find variance of predictability on the observed scale, $\sigma_{\sigma_w^2}^2$. To do this, we enter the ln-normal scale values of the total variance in predictability, $\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2$, and the average residual variance, $\beta_{pv0,exp}$ into the formula for converting variance from a ln-normal distribution (Equation 37), such that:

$$\sigma_{\sigma_w^2}^2 = \left(\exp\left(\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2\right) - 1\right) \exp\left(2\beta_{pv0,exp} + \sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2\right), \quad (42)$$

Then, the formula for $\sigma_{\sigma_p^2}^2$ is provided by Mulder et al. (2007) as:

$$\sigma_{\sigma_p^2}^2 = 2\sigma_p^4 + 3\sigma_{\sigma_w^2}^2, \quad (43)$$

where the value for σ_p^2 is shown in Equation 40.

4.2.5 | Between-individual variance for predictability

In our case, the between-individual variance for predictability is $\sigma_{ID_v}^2 = \sigma_{ID_{v0}}^2$, so we need to convert $\sigma_{ID_{v0,exp}}^2$ (from the ln-normal scale) to $\sigma_{ID_{v0}}^2$. Our first thought might be to apply the same transformation to $\sigma_{ID_{v0,exp}}^2$ as we did for $\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2$ (i.e. Equation 37). However, because the ln-transformation is nonlinear, we cannot simply disentangle $\sigma_{ID_{v0,exp}}^2$ from $\sigma_{fixed_{v,exp}}^2$. The solution, provided by Mulder et al. (2007), is to assume that the proportionality of

variance components is preserved across different scales (see also Sae-Lim et al., 2015) so that:

$$\sigma_{ID_{v0}}^2 = \sigma_w^2 \left(\frac{\sigma_{ID_{v0,exp}}^2}{\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2} \right), \quad (44)$$

where σ_w^2 was calculated in Equation 42 (on the observed scale, we can write $\sigma_w^2 = \sigma_{ID_{v0}}^2 + \sigma_{fixed_v}^2$). Thus, we are assuming the ratio of variance components on the ln-normal scale is the same as the ratio of variance components on the observed scale: $\frac{\sigma_{ID_{v0,exp}}^2}{\sigma_{ID_{v0,exp}}^2 + \sigma_{fixed_{v,exp}}^2} = \frac{\sigma_{ID_{v0}}^2}{\sigma_{ID_{v0}}^2 + \sigma_{fixed_v}^2}$ (we refer to this assumption as 'the preservation of proportionality').

4.3 | Comparing estimates between studies

When standardising variance estimates, it is important to consider the scale of measurement, whether or not data were transformed prior to analysis, and mean–variance relationships (e.g. comparing CV_{ID} across traits becomes challenging when mean–variance relationships deviate from proportionality predicted by Taylor's law). An accessible summary of the limitations of coefficients of variation is provided by Hansen et al. (2011) and Pélabon et al. (2020).

Between-study comparisons of the magnitude of individual differences would ideally re-analyse the raw data from original studies (which are increasingly made publicly available by authors in ecology and evolution). In addition to providing raw data, when reporting the results of DHGLMs, we recommend authors report all variance components (including the fixed effect variance), as well as the population intercept for the dispersion model. Standardising the way R_p and CV_{ID} are calculated is important because between-study variance in estimates can be increased by variation in statistical methods and chosen formulas (e.g. was fixed effect variance included or excluded from the total phenotypic variance?). Calculating R_p and CV_{ID} from scratch also allows sampling variance to be estimated for meta-analytic models.

In addition to being influenced by analysis decisions, R_p and CV_{ID} can vary due to different experimental and sampling designs (Wilson, 2018). For instance, a statistical difference between individuals could reflect the effects of measuring individuals in different conditions (e.g. due to being sampled at different times), rather than true between-individual differences (e.g. 'pseudo-repeatability'; Dingemanse & Dochtermann, 2013). Likewise, a short sampling interval between repeated measurements is likely to inflate estimates of individual differences, due to temporal autocorrelation. It is also important to consider the impact that sampling intervals have on individual's behavioural responses (e.g. habituation) and, within studies, standardise these intervals across individuals.

For comparisons of CV_{ID} , two additional points are important to consider. First, were data transformed prior to analysis? If so, estimated parameters need to be brought back to the observed scale (this applies both to comparisons across studies, and comparisons within studies for different phenotypic traits). The supplementary worked example describes how to reverse linear transformations

(e.g. z-scaling) and nonlinear transformations (e.g. log- or square-root transformations, which are commonly done to improve the normality of residuals. For a DHGLM, violations of normality cause problems with the estimation of variance in predictability). Second, when comparing estimates of CV_{ID_v} to another study, did that study also use residual variances as the response variable for the dispersion model, $\ln(\sigma_{e_{ij}}^2)$, or did it use residual standard deviations, $\ln(\sigma_{e_{ij}})$, as in Cleasby et al. (2015) and the current default in the *R* package *BRMS* (Bürkner, 2018)? Parameters from the dispersion models can be converted between these two scales using the relationship $\frac{1}{2} \ln(\sigma_{e_{ij}}^2) = \ln(\sigma_{e_{ij}})$ (more details are provided in the Supporting Information, including equations for converting between CV_{ID_v} and $CV_{ID_{sd}}$). See the supplementary *R* code (O'Dea et al., 2021) for conversions between the $\ln(\sigma_{e_{ij}}^2)$ and $\ln(\sigma_{e_{ij}})$ models, fit with the *RSTAN* (v. 2.21.2; Stan Development Team, 2020) and *BRMS* (v. 2.15.0; Bürkner, 2018) packages, respectively.

5 | CONCLUSIONS AND FUTURE DIRECTIONS

Incorporating predictability into studies of personality and plasticity creates an opportunity to test more nuanced questions about how phenotypic variation is maintained, or constrained. For some traits, it might be adaptive to be unpredictable, such as in predator–prey interactions (Briffa, 2013). For other traits, selection might act to minimise maladaptive imprecision around an optimal mean (Hansen et al., 2006). The supplementary worked example and open code (O'Dea et al., 2021) shows between-individual correlations in predictability across multiple behavioural traits, and some correlations of predictability with personality and plasticity. If driven by biological integration and not measurement errors or statistical artefacts, these correlations could hint at genetic integration too; other studies have found additive genetic variance in predictability (Martin et al., 2017; Prentice et al., 2020). Given that different traits might have different optimal levels of unpredictability, integration of predictability could constrain variation in one trait (resulting in lower than optimal variability) and maintain variation in another (resulting in greater than optimal variability). Because of associations with personality and plasticity, variation in predictability—the lowest level of the phenotypic hierarchy—could have cascading effects upwards (Westneat et al., 2015). Empirical estimates of the strength of these associations can inform theoretical models on the simultaneous evolution of means and variances.

5.1 | Beyond behaviour

We focussed this paper on animal behaviour (the field we are most familiar with), but the models are broadly adaptable. Individuals can show differences in predictability for any trait

that is repeatedly expressed. For example, medical researchers might want to quantify the variability of patient's drug responses (Nettles et al., 2006), and selective breeders of plants might want to reduce individual variability in seed or fruit mass (Herrera, 2017). The review by Herrera (2017) discusses the overlooked importance of variability within the structures of an individual plant, including for plant–animal interactions. Given the large sample sizes required to estimate multiple individual differences, the most tractable tests of the synchronous evolution of means and variances could come from non-animal systems. Clonal species can also be used to estimate individual differences in predictability of non-labile traits.

6 | CONCLUSIONS

While many studies quantify consistent individual differences in repeatedly expressed traits, such as behaviour, much of the mystery of phenotypic variation is obscured within residual variation. Individuals impact the world not only through their 'average' phenotype, but also through their extremes. Given that evolution can act on both averages and variances, to understand the evolution of labile traits, we need to measure both the magnitude and consistency of individual differences, as well as their associations. Limitations of the concepts and tools presented here include difficulties differentiating biological integration from correlations driven by measurement or design errors, the high sample sizes required to accurately estimate variance components and covariances, and concerns about inflated rates of false-positive findings when estimating many parameters. Future simulation work is required to help empiricists design adequate sampling methods to chronicle the integration of multiple levels of phenotypic variation in diverse systems. In doing so, we can improve our understanding of the factors promoting and constraining variability, as well as the evolution, and ecological consequences, of individuality.

ACKNOWLEDGEMENTS

This work was conducted with permission from the Garvan Institute Animal Ethics Committee (approval number 15/15) with funding support from UNSW. Daniel Hesselton provided access to the zebrafish population featured in the supplementary worked example, which were looked after by staff within the Garvan Institute of Medical Research. Data were collected with help from Malgorzata Lagisz, Melissa Fangmeier, Takuji Usui, Marissa Baptista, Alex Lincoln-Dodgson, Hamza Anwer, Harry Thomas and Alexander Aloy. Joel Pick helped write the code for bivariate predictability models. R.E.O. was supported by an Australian Government Research Training Program Scholarship, D.W.A.N. was supported by an ARC DECRA fellowship (DE150101774), S.N. was supported by an ARC Discovery grant (DP180100818). We would like to thank Bob Wong, Alison Bell, Julien Martin and five anonymous reviewers for constructive comments on earlier versions of this manuscript.

CONFLICT OF INTEREST

All authors declare no conflicts of interest.

AUTHORS' CONTRIBUTIONS

R.E.O. conceptualisation, data curation, formal analysis, investigation, methodology, project administration, software, visualisation, writing—original draft, writing—reviewing and editing. D.W.A.N. conceptualisation, investigation, methodology, resources, software, writing—reviewing and editing. S.N. conceptualisation, investigation, methodology, resources, software, supervision, writing—reviewing and editing.

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1111/2041-210X.13755>.

DATA AVAILABILITY STATEMENT

The statistical models described in this review are demonstrated in a supplementary worked example, which can be reproduced using the data, code and model objects contained in this dedicated repository: <http://doi.org/10.17605/OSF.IO/V3QAX> (O'Dea et al., 2021).

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How to cite this article: O'Dea, R. E., Noble, D. W. A., & Nakagawa, S. (2021). Unifying individual differences in personality, predictability and plasticity: A practical guide. *Methods in Ecology and Evolution*, 00, 1–16. <https://doi.org/10.1111/2041-210X.13755>